

Qualification Pack



Technician - TQC for Plastic Materials & Products

QP Code: RSC/Q0420

Version: 3.0

NSQF Level: 4

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RSC/Q0420: Technician - TQC for Plastic Materials & Products

Brief Job Description

The Technician - TQC for Plastic Materials & Products ensures quality compliance by conducting physical, mechanical, and non-destructive tests on plastic raw materials and finished goods. The role involves operating and calibrating testing equipment, identifying defects, and maintaining quality records as per BIS/ISO standards. The technician supports production and QA teams in ensuring product reliability and customer satisfaction.

Personal Attributes

The job requires the individual to have good observational skills, with an eye to detail, and positive attitude.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N0408: Conduct Testing of Plastic Raw Materials and Finished Products](#)
2. [RSC/N0409: Ensure Compliance with Quality Standards](#)
3. [RSC/N0411: Calibrate and Maintain Testing Equipment](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Regulatory Affairs and Compliance
Occupation	Testing and Quality Control
Country	India
NSQF Level	4
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2141.0100

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Minimum Educational Qualification & Experience	12th grade Pass (or equivalent) OR 10th grade pass plus 1-year NTC/ NAC with 1.5 years of experience Relevant OR 10th grade pass with 3 Years of experience Relevant OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	3.0
Reference code on NQR	QG-04-RI-04047-2025-V2-RSDC
NQR Version	2

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RSC/N0408: Conduct Testing of Plastic Raw Materials and Finished Products

Description

This NOS covers the skills and knowledge required to carry out a wide range of tests on plastic raw materials and finished products using destructive and non-destructive techniques. It includes specimen preparation, test execution, analysis of results, and proper documentation.

Scope

The scope covers the following :

- Prepare for Testing
- Conduct Physical and Mechanical Tests
- Perform Non-Destructive Testing
- Analyze and Report Results

Elements and Performance Criteria

Prepare for Testing

To be competent, the user/individual on the job must be able to:

- PC1.** identify material type and testing requirements from work order
- PC2.** select suitable test methods based on product and standards
- PC3.** collect and label raw material/product samples properly
- PC4.** prepare specimens using standard cutting/molding tools
- PC5.** check testing instruments for functionality and cleanliness
- PC6.** set up environmental conditions as per test requirements

Conduct Physical and Mechanical Tests

To be competent, the user/individual on the job must be able to:

- PC7.** perform MFI (Melt Flow Index) tests
- PC8.** conduct tensile, elongation, and modulus testing
- PC9.** carry out impact testing (Izod/Charpy)
- PC10.** execute hardness tests (Shore A/D, Rockwell)
- PC11.** measure HDT/Vicat softening point
- PC12.** conduct specific gravity/density tests
- PC13.** test for opacity, gloss, haze, and color (if applicable)
- PC14.** log all test parameters and results systematically

Perform Non-Destructive Testing

To be competent, the user/individual on the job must be able to:

- PC15.** identify components requiring NDT based on specifications
- PC16.** perform visual inspection for surface defects
- PC17.** use ultrasonic testing (UT) to detect internal flaws (where applicable)

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- PC18.** apply infrared thermography to detect non-uniformities
- PC19.** conduct leak testing using bubble or vacuum method
- PC20.** document findings with supporting images or data
- PC21.** handle NDT equipment as per calibration and safety norms

Analyze and Report Results

To be competent, the user/individual on the job must be able to:

- PC22.** calculate average values and analyze deviations
- PC23.** compare results with specification or tolerance limits
- PC24.** flag out-of-spec data and initiate retesting or reporting
- PC25.** document all findings in lab records or software
- PC26.** communicate any abnormalities to QA or production team

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** classification and characteristics of plastics
- KU2.** purpose and process of MFI, tensile, impact, HDT, hardness tests
- KU3.** importance of test specimen conditioning
- KU4.** tolerances and standard deviations in results
- KU5.** test equipment types, functions, and operating principles
- KU6.** standards (ASTM, ISO, BIS) used in plastic testing
- KU7.** common errors and corrective actions in testing
- KU8.** safety protocols in testing environments
- KU9.** recordkeeping and traceability in labs
- KU10.** overview of Non-Destructive Testing in plastics
- KU11.** types of NDT: Visual, Ultrasonic, Infrared, Leak testing
- KU12.** applications and limitations of NDT
- KU13.** interpretation of NDT findings and documentation
- KU14.** maintenance of NDT tools

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret occupational health and safety (OHS) standards, legislative requirements, and industry standards
- GS2.** communication effectively with colleagues, subordinates and seniors
- GS3.** note down the problem for identification of the appropriate solution

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for Testing</i>	6	12	-	-
PC1. identify material type and testing requirements from work order	1	2	-	-
PC2. select suitable test methods based on product and standards	1	2	-	-
PC3. collect and label raw material/product samples properly	1	2	-	-
PC4. prepare specimens using standard cutting/molding tools	1	2	-	-
PC5. check testing instruments for functionality and cleanliness	1	2	-	-
PC6. set up environmental conditions as per test requirements	1	2	-	-
<i>Conduct Physical and Mechanical Tests</i>	11	27	-	-
PC7. perform MFI (Melt Flow Index) tests	1	3	-	-
PC8. conduct tensile, elongation, and modulus testing	2	4	-	-
PC9. carry out impact testing (Izod/Charpy)	1	3	-	-
PC10. execute hardness tests (Shore A/D, Rockwell)	2	4	-	-
PC11. measure HDT/Vicat softening point	2	4	-	-
PC12. conduct specific gravity/density tests	1	3	-	-
PC13. test for opacity, gloss, haze, and color (if applicable)	1	3	-	-
PC14. log all test parameters and results systematically	1	3	-	-
<i>Perform Non-Destructive Testing</i>	9	23	-	-
PC15. identify components requiring NDT based on specifications	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC16. perform visual inspection for surface defects	1	3	-	-
PC17. use ultrasonic testing (UT) to detect internal flaws (where applicable)	2	4	-	-
PC18. apply infrared thermography to detect non-uniformities	1	3	-	-
PC19. conduct leak testing using bubble or vacuum method	2	4	-	-
PC20. document findings with supporting images or data	1	3	-	-
PC21. handle NDT equipment as per calibration and safety norms	1	3	-	-
<i>Analyze and Report Results</i>	4	8	-	-
PC22. calculate average values and analyze deviations	1	2	-	-
PC23. compare results with specification or tolerance limits	1	2	-	-
PC24. flag out-of-spec data and initiate retesting or reporting	1	2	-	-
PC25. document all findings in lab records or software	1	2	-	-
PC26. communicate any abnormalities to QA or production team	1	2	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N0408
NOS Name	Conduct Testing of Plastic Raw Materials and Finished Products
Sector	Rubber
Sub-Sector	Plastic Manufacturing
Occupation	Quality Control
NSQF Level	4
Credits	6
Version	2.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N0409: Ensure Compliance with Quality Standards

Description

This OS unit is about monitoring and ensuring quality compliance during all stages - raw material inspection, in-process checks, and finished product verification. It includes defect identification, data analysis, documentation, and audit readiness.

Scope

The scope covers the following :

- Inspect Incoming Raw Materials
- Monitor In-Process Quality
- Verify Finished Product Conformance
- Support Quality Documentation and Audits

Elements and Performance Criteria

Inspect Incoming Raw Materials

To be competent, the user/individual on the job must be able to:

- PC1.** review Certificate of Analysis (CoA) and match with specifications
- PC2.** conduct visual and basic physical tests
- PC3.** document acceptance or rejection with reasons

Monitor In-Process Quality

To be competent, the user/individual on the job must be able to:

- PC4.** perform hourly sampling and testing from the line
- PC5.** check process parameters and match with control plan
- PC6.** identify visual and dimensional defects during production
- PC7.** escalate non-conformances to supervisors
- PC8.** record production quality data

Verify Finished Product Conformance

To be competent, the user/individual on the job must be able to:

- PC9.** perform final Quality Control (QC) tests (strength, dimensions, aesthetics)
- PC10.** tag and segregate accepted and rejected lots
- PC11.** conduct NDT where required on finished products
- PC12.** re-inspect post-repair items using NDT

Support Quality Documentation and Audits

To be competent, the user/individual on the job must be able to:

- PC13.** maintain inspection checklists and batch records
- PC14.** help prepare Corrective And Preventive Actions (CAPA) documents
- PC15.** support internal/external audit teams

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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** BIS/ISO quality standards for plastic products
- KU2.** control plans, CoAs, and inspection criteria
- KU3.** Statistical Process Control (SPC), CP/CPK basics
- KU4.** visual inspection methods and AQL
- KU5.** common manufacturing defects (sink marks, flash, warpage)
- KU6.** traceability and batch-level documentation
- KU7.** defect classification: minor, major, critical
- KU8.** QC tools (7): Pareto, Fishbone, Histogram, etc.
- KU9.** basics of root cause analysis and CAPA
- KU10.** integration of NDT findings in QA processes

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret GMP, ASTM Standards, FMEAs) and other analysis
- GS2.** write or prepare inspection and test reports
- GS3.** communication effectively with colleagues, subordinates and seniors
- GS4.** escalate the problem in an appropriate manner
- GS5.** analyze the problem and identify appropriate solution

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect Incoming Raw Materials</i>	2	9	-	-
PC1. review Certificate of Analysis (CoA) and match with specifications	1	3	-	-
PC2. conduct visual and basic physical tests	1	3	-	-
PC3. document acceptance or rejection with reasons	1	3	-	-
<i>Monitor In-Process Quality</i>	14	28	-	-
PC4. perform hourly sampling and testing from the line	3	6	-	-
PC5. check process parameters and match with control plan	3	6	-	-
PC6. identify visual and dimensional defects during production	3	6	-	-
PC7. escalate non-conformances to supervisors	3	6	-	-
PC8. record production quality data	2	4	-	-
<i>Verify Finished Product Conformance</i>	12	24	-	-
PC9. perform final Quality Control (QC) tests (strength, dimensions, aesthetics)	3	6	-	-
PC10. tag and segregate accepted and rejected lots	3	6	-	-
PC11. conduct NDT where required on finished products	3	6	-	-
PC12. re-inspect post-repair items using NDT	3	6	-	-
<i>Support Quality Documentation and Audits</i>	2	9	-	-
PC13. maintain inspection checklists and batch records	1	3	-	-
PC14. help prepare Corrective And Preventive Actions (CAPA) documents	1	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. support internal/external audit teams	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N0409
NOS Name	Ensure Compliance with Quality Standards
Sector	Rubber
Sub-Sector	Plastic Manufacturing
Occupation	Quality Control
NSQF Level	4
Credits	4
Version	2.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N0411: Calibrate and Maintain Testing Equipment

Description

This OS unit is about periodic calibration, troubleshooting, and maintenance of lab and testing equipment, including both destructive and non-destructive test instruments. It ensures reliability and accuracy in test results.

Scope

The scope covers the following :

- Plan Calibration Activities
- Perform Calibration
- Maintain Equipment and Records

Elements and Performance Criteria

Plan Calibration Activities

To be competent, the user/individual on the job must be able to:

- PC1.** identify instruments needing calibration
- PC2.** refer to SOPs for schedule and frequency
- PC3.** coordinate with external vendors if applicable

Perform Calibration

To be competent, the user/individual on the job must be able to:

- PC4.** use certified standard tools and materials
- PC5.** compare readings against reference standards
- PC6.** adjust/calibrate instruments as needed
- PC7.** tag calibrated instruments with updated info
- PC8.** calibrate NDT tools (e.g., infrared camera, ultrasonic testers)

Maintain Equipment and Records

To be competent, the user/individual on the job must be able to:

- PC9.** clean and store tools properly
- PC10.** troubleshoot common equipment faults
- PC11.** replace worn-out components
- PC12.** maintain calibration history sheets
- PC13.** schedule preventive maintenance tasks

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of equipment calibration in quality systems
- KU2.** calibration techniques for MFI, tensile testers, UTM, etc.

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- KU3.** concepts of accuracy, repeatability, and error margins
- KU4.** SOPs for preventive and corrective maintenance
- KU5.** handling and storage of sensitive instruments
- KU6.** calibration of NDT tools (ultrasonic, IR thermography)
- KU7.** use of software for calibration logs and alerts
- KU8.** equipment tagging and audit-readiness

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret GMP, blueprints, data, manuals, or other materials to determine specifications, inspection and testing procedures, adjustment methods, certification processes, formulas, or measuring instruments required
- GS2.** communication effectively with upstream and downstream teams
- GS3.** note down the problem for identification of appropriate solution

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Plan Calibration Activities</i>	6	12	-	-
PC1. identify instruments needing calibration	2	4	-	-
PC2. refer to SOPs for schedule and frequency	2	4	-	-
PC3. coordinate with external vendors if applicable	2	4	-	-
<i>Perform Calibration</i>	16	38	-	-
PC4. use certified standard tools and materials	2	4	-	-
PC5. compare readings against reference standards	3	7	-	-
PC6. adjust/calibrate instruments as needed	4	10	-	-
PC7. tag calibrated instruments with updated info	3	7	-	-
PC8. calibrate NDT tools (e.g., infrared camera, ultrasonic testers)	4	10	-	-
<i>Maintain Equipment and Records</i>	8	20	-	-
PC9. clean and store tools properly	2	4	-	-
PC10. troubleshoot common equipment faults	2	4	-	-
PC11. replace worn-out components	2	4	-	-
PC12. maintain calibration history sheets	2	4	-	-
PC13. schedule preventive maintenance tasks	2	4	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N0411
NOS Name	Calibrate and Maintain Testing Equipment
Sector	Rubber
Sub-Sector	Plastic Manufacturing
Occupation	Quality Control
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

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- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

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- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N0408. Conduct Testing of Plastic Raw Materials and Finished Products	30	70	-	-	100	30
RSC/N0409. Ensure Compliance with Quality Standards	30	70	-	-	100	30
RSC/N0411. Calibrate and Maintain Testing Equipment	30	70	-	-	100	20
RSC/N5007. Follow health & safety protocol	30	70	-	-	100	10
DGT/VSQ/N0101. Employability Skills (30 Hours)	20	30	-	-	50	10
Total	140	310	-	-	450	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.